Data Matrix Analysis

AP

Motivations

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Activity tables show how users *map* their choices,

or how available products *map* onto their adopters.

	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

Figure 11.6: Ratings of movies by users

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Figure 11.6: Ratings of movies by users

Data Science view: 7 user profiles; 18 datapoints of user review.

Network science view: a weighted, binary relationship between users and films.

Let's step back and see what Geometry gives us.

Spectral Methods

Most activity matrices represent the connections between n entities (e.g., subscribers) and m entities (e.g., films) with $n <> m$.

Sometimes the connections is between the same entities, such as endorsement, teams defeating other teams, friends or followers on social networks etc.

In such cases, the matrix is *square*.

Standard Geometry holds and we can extract its *eigenpairs*.

Matrices

Traditional view

There is a *space* in n dimensions

Each point is represented by an n-dimensional vector $\mathbf{x}$ (or $\vec{x}$)

Transformation: a *expansion* or a *contraction,* possibly combined with a *rotation,* is represented by a matrix, A

The resulting point is computed as $A\mathbf{x}$.

Example

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

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$$M\mathbf{x} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

M does anti-clockwise rotation and expansion at once.

More examples

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expand $\mathbf{x}$ to double its length:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

compress $\mathbf{x}$ to halve its length:

$$\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.25 \end{pmatrix}$$

rotate $\mathbf{x}$ anticlockwise:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 1 \end{pmatrix}$$

rotate $\mathbf{x}$ clockwise:

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -1 \end{pmatrix}$$

change sense:

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} -1 \\ -0.5 \end{pmatrix}$$

Eigenpairs

Definition

If matrix A has a real λ and a vector $\mathbf{e}$ s.t.

$$A\mathbf{e} = \lambda\mathbf{e}$$

then λ is an *eigenvalue* and $\mathbf{e}$ an *eigenvector* of A.

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If A has rank n then there could be up to n eigenpairs.
In practice,

- they might not be real, nor $\neq 0$, and
- are always *costly*: n^{α}) scalar multiplications, n being the size of the matrix and $\alpha \approx 2.81$.

Eigendecomposition

The *effect* of A can be decomposed into translation + rotation thanks to the extraction of *eigenpairs* $\langle \boldsymbol{\lambda}, \mathbf{e} \rangle$:

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Consider the M matrix again:

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

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solve $M\mathbf{e} = \lambda\mathbf{e}$ to find

$\lambda_1 \approx 5.372$ and $\lambda_2 \approx -0.372$.

$$\lambda_1 \approx 5.372 \text{ and } \lambda_2 \approx -0.372.$$

$$\mathbf{e_1} \approx \begin{pmatrix} 0.416 \\ 0.909 \end{pmatrix}$$

$$\mathbf{e_2} \approx \begin{pmatrix} 0.824 \\ -0.0566 \end{pmatrix}$$

M now can be re-defined as the product of simpler operations:

$$M = Q\Lambda Q^{-1}$$

Stretching, followed by rotation followed by 'stretching back'

$$Q = \begin{pmatrix} \mathbf{e_1} & \mathbf{e_2} \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Finally

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} =$$

$$\begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.575 & 0 \\ 0.924 & -0 \end{pmatrix}$$

Example application

$$M\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

$$\begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.575 & 0 \\ 0.924 & -0 \end{pmatrix}$$

$$M\mathbf{x} = \begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.372 & 0 \\ 0 & -0.372 \end{pmatrix} \begin{pmatrix} 0.9 \\ 0.7 \end{pmatrix}$$

$$M\mathbf{x} = \begin{pmatrix} 0.416 & 0.824 \\ 0.909 & -0.566 \end{pmatrix} \begin{pmatrix} 5.340 \\ -0.265 \end{pmatrix}$$

$$M\mathbf{x} = \begin{pmatrix} 2.003 \\ 5.004 \end{pmatrix}$$

Interesting square matrices

A is called *symmetric* when $A = A^T$

Example matrix M is not symmetric:

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = M^T$$

A is called *positive semidefinite* when, for any $\mathbf{x}$, we have

$$\mathbf{x}^T A \mathbf{x} \geq 0$$

In such case its eigenvalues are guaranteed non-negative: $\lambda_i \geq 0$,

this helps interpretation of activity matrices.

Singular Value Decomposition

For rectangular matrices Eigendecomposition is not defined.

SVD will provide a similar decomposition for data matrices of any size.

Details are in the [Goodfellow et al.] textbook and will be introduced later on.

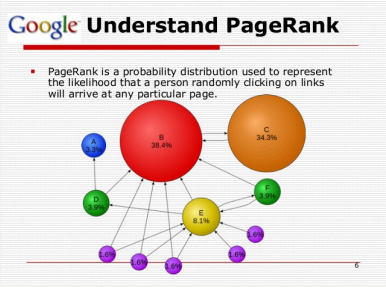
Applications

Spectral properties

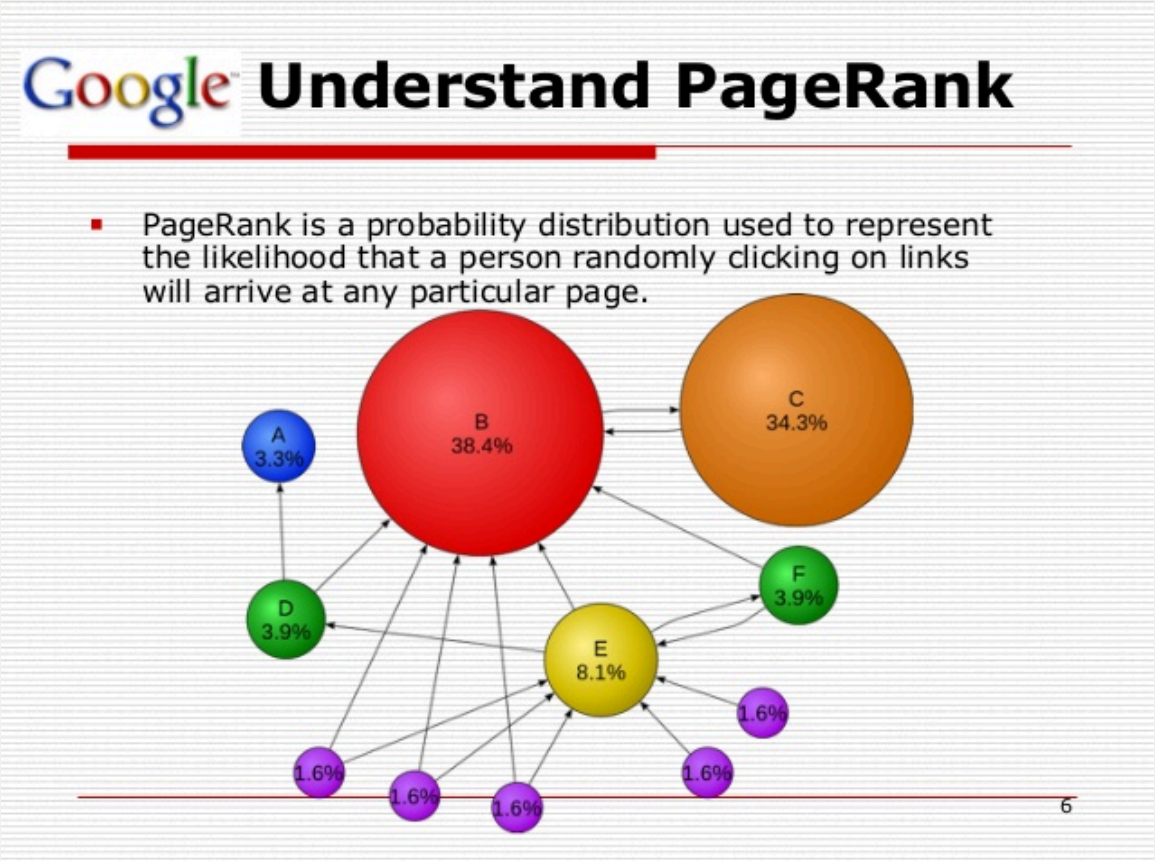
Adjacency matrices represent connections between entities in a network (graph), e,g., the Web.

The eigenvalues of adjacency matrices provide bounds for several network features.

Google PageRank algorithm *is* spectral network anal.



Web page importance ≈ eigenvector centrality:



Early applications in Psychology, Social science, Bibliometrics, Economics, and Choice theory

(seriously).

Spectral ranking

Given a matrix representing preference or likeability between people, can we rank the participants (from best to worst) on the basis of their general, intrinsic likeability?

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my *likeability index* should be equal to the weighted sum of of the indices of the people who like me.

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But their likeability is turn will depend on mine…

Let's use row vectors $\mathbf{r} = [r_1, r_2, \ldots r_n]$:

$$\mathbf{r} = \mathbf{r}M$$

i.e., $\mathbf{r}$ is a left eigenvector of M.

This formula might have no solution, but matrix preprocessing can assure that one exists.

Study plan

Background study

Ian Goodfellow, Yoshua Bengio and Aaron Courville:
Deep Learning, MIT Press, 2016.

available in HTML and PDF; it is *a refresher* of notation and properties: no examples and no exercises.

It can be read in the background.

- Phase 1: read §§ 2.1—2.7, then § 2.11.
- Phase 2: read §§ 2.8—2.10